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SUMMARY

A doctoral researcher at Purdue University focused on advanced control of building HVAC systems. Studying reinforcement learning, model predictive control, and data-driven predictive control for buildings to improve energy efficiency, reduce costs and emissions, and support scalable deployment. This work emphasizes practical deployment through experiment protocols and scalable controller design. Previously earned an MS in Mechanical Engineering from the University of British Columbia, Canada, and interned at Tesla, where I worked on machine learning for quality assessment.

EDUCATION

Purdue University, West Lafayette, IN PhD in Mechanical Engineering Specialization in Computational Science and Engineering	08/2023 – Present GPA: 3.9/4.0
University of British Columbia, BC, Canada MS in Mechanical Engineering	05/2021 – 08/2023 GPA: 94% (4.0/4.0)
University of Tabriz, EA, Iran BS in Mechanical Engineering	08/2016 – 07/2020 GPA: 19.12/20 (4.0/4.0) – Rank: 1/155+

PUBLICATIONS

Journal Articles

2. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, P. Papageorgiou, A.H. Lee, J.E. Braun, G.P. Henze, and K.J. Kircher, “Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: A review.”
Applied Energy, 2025.
1. **A.J. Khabbazi**, M. Zabihi, R. Li, M. Hill, V. Chou, and J. Quinn, “Mixing hydrogen into natural gas distribution pipeline system through Tee junctions.”
International Journal of Hydrogen Energy, 2024.

Conference Papers

5. **A.J. Khabbazi** and K.J. Kircher, “Small HVAC Control Demonstrations in Larger Buildings Often Overestimate Savings.”
American Control Conference (ACC), 2026.
4. E.N. Pergantis, L.D. Reyes Premer, **A.J. Khabbazi**, Priyadarshan, F. Wu, D. Ziviani, and K.J. Kircher, “Active current limiting control of residential appliances for breaker panel protection across the US: a preliminary study.”
CISBAT, 2025.
3. **A.J. Khabbazi**, “PhD Forum Abstract: Field Demonstration of Advanced Commercial HVAC Control for Scalable Deployment.”
ACM BuildSys, 2025.
2. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, A.H. Lee, J. Ma, H. Liu, G.P. Henze, and K.J. Kircher, “What Have We Learned From Field Demonstrations of Advanced Commercial HVAC Control?”
International High Performance Buildings Conference, 2024.
1. **A.J. Khabbazi**, M. Zabihi, R. Li, V. Chou, and J. Quinn, “Blending of Hydrogen into a Natural Gas Distribution Pipeline in British Columbia through a Tee Junction for Reducing GHG Emissions.”
Canadian Society for Mechanical Engineering (CSME) International Congress, 2023.

HONORS & AWARDS

ASHRAE Graduate Student Grant-in-Aid Award	2025
Best PhD Forum Presentation Runner-up Award, ACM BuildSys	2025
Best Paper Award, CSME International Congress	2023
Best Presentation Award, CSME International Congress (Advanced Energy Symposium)	2022
UBC Graduate Scholarship, University of British Columbia	2022
UBC Dean’s Entrance Scholarship, University of British Columbia	2021

INDUSTRY EXPERIENCE

Tesla, Inc., Summer Intern, Applied ML, Sparks, NV

05/2025–08/2025

Machine Learning for Defect Detection and Quality Assessment of Drive Units

- Built supervised ML models (XGBoost, DNNs, SVR, and ensembles) for defect detection and quality assessment.
- Developed data pipelines in MySQL and engineered features to extract high-signal inputs for ML models.
- Contributed to deployment of a thermal imaging system for non-destructive inspection in drive-unit manufacturing.
- Collaborated on projects projected to reduce testing costs by over \$1M/year through improved quality assessment.

RESEARCH EXPERIENCE

Purdue University, Graduate Research Assistant (PhD), West Lafayette, IN

08/2023–Present

Advanced HVAC Control in Real Buildings: State of the Art, Protocols, and Deployment

- Derived a multi-zone formulation for HVAC control savings overestimation from cross-zone heat transfer (*ACC26*).
- Reviewed 100+ publications on MPC and RL for HVAC, resulting in a [review paper](#); [GitHub](#).
- Helped develop active current-limiting control for appliances to prevent breaker overloads in electrified homes.
- Developed an RL controller for residential HVAC using OCHRE, saving 17.5% energy while maintaining comfort.
- Designed data-enabled predictive control (DeePC) for HVAC, matching MPC without an explicit system model.
- Supported commissioning of a university building by upgrading a Niagara-based building automation system.
- Created real-time dashboards for energy/comfort and ML models for occupancy detection (98% accuracy).

University of British Columbia, Graduate Research Assistant (MS), BC, Canada

05/2021–08/2023

Mixing Hydrogen into Natural Gas Distribution Pipeline System Through Tee Junctions

- Performed CFD simulations and developed ideal- and real-gas mixture models in C; [GitHub](#).
- Worked with **FortisBC Energy Inc.** research teams to assess industry impacts as part of [UBC H₂Lab](#) projects.
- Produced one [journal paper](#), three conference papers, and an [MS thesis](#), earning two awards.

TEACHING EXPERIENCE

Purdue University, Graduate Teaching Assistant (PhD)

- ME597: Distributed Energy Resources (DERs)

Spring'26

University of British Columbia, Graduate Teaching Assistant (MS)

- Engineering Analysis I
- Fluid Mechanics II Lab

Fall'21, Spring'22, Fall'22

Spring'22, Spring'23

SELECTED COURSES

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|---|---|
| • ECE570: Artificial Intelligence | • STAT511: Statistical Methods |
| • IE690: Reinforcement Learning & Control (RLC) | • ME597: Industrial Internet of Things (IIoT) |
| • AAE568: Applied Optimal Control & Estimation | • ME597: Distributed Energy Resources (DER) |
| • AAE 561: Intro to Convex Optimization | • ME511: Analysis of Thermal Systems |

SKILLS

Expertise: Predictive Control, Reinforcement Learning, Data Structures & Algorithms, Convex Optimization, Control Theory, Machine Learning, Deep Learning, Generative AI, Industrial IoT, Buildings & HVAC, Thermodynamics.

Programming & Tools: Python, MATLAB, C, Git, Linux, Docker.

ML & Optimization: PyTorch, TensorFlow, scikit-learn, XGBoost, Stable-Baselines3, CVX, CVXPY, Gurobi

Data Analysis & Visualization: Pandas, NumPy, SciPy, Matplotlib, Seaborn, Grafana.

Simulation & Testbeds: Gymnasium, OCHRE, BOPTEST.

Databases & Integration: MySQL (SQL), InfluxDB, Tridium Niagara, Modbus, IO-Link, MTConnect.